Fundamental Concepts

(1) Electric field of a point charge
(2) Relationship between electric field and electric force
(2) Magnetic field of a moving point charge
(4) Relationship between magnetic field and magnetic force on a point charge

Conservation of Charge The Superposition Principle Conservation of Energy

The Momentum Principle (Newton's 2d law): $\frac{d\vec{p}}{dt} = \vec{F}_{net}$ or $m\vec{a} = \vec{F}_{net}$

$$\oint \vec{E} \bullet \hat{n}dA \approx \sum \vec{E} \bullet \hat{n}\Delta A = \frac{\sum q_{inside}}{\varepsilon_0} \qquad \oint \vec{B} \bullet \hat{n}dA \approx \sum \vec{B} \bullet \hat{n}\Delta A = 0 \qquad \oint \vec{B} \bullet d\vec{l} = \mu_0\left[\sum I_{inside\ path} + \varepsilon_0\frac{d}{dt}\int \vec{E} \bullet \hat{n}dA\right]$$

$$\Phi_{mag} = \int \vec{B} \bullet \hat{n}dA \qquad \oint \vec{E} \bullet d\vec{l} = emf \qquad |emf| = \left|N\frac{d\Phi_{mag}}{dt}\right| \qquad \oint \vec{E} \bullet d\vec{l} = -\frac{d}{dt}\int \vec{B} \bullet \hat{n}dA$$

$$\Delta V = V_f - V_i = -\int_i^f \vec{E} \bullet d\vec{l} \approx -\sum \vec{E} \bullet \Delta\vec{l} \qquad q\Delta V = \Delta U$$

Specific results

Electric field of a uniformly charged spherical shell: outside shell, like point charge; inside shell, 0.

$\left|\vec{E}_{rod}\right| = \frac{1}{4\pi\varepsilon_0}\frac{Q}{r\sqrt{r^2 + L/2^2}}$ a $\perp$ distance r from center $\left|\vec{E}_{rod}\right| \approx \frac{1}{4\pi\varepsilon_0}\frac{2(Q/L)}{r}$ if $r << L$ $\left|\vec{E}_{ring}\right| = \frac{1}{4\pi\varepsilon_0}\frac{qz}{(z^2 + R^2)^{3/2}}$ on axis

$\left|\vec{E}_{disk}\right| = \frac{Q/A}{2\varepsilon_0}\left[1 - \frac{z}{(z^2 + R^2)^{1/2}}\right]$ $\left|\vec{E}_{disk}\right| \approx \frac{Q/A}{2\varepsilon_0}\left[1 - \frac{z}{R}\right] \approx \frac{Q/A}{2\varepsilon_0}$ if $z << R$

$\left|\vec{E}_{capacitor}\right| \approx \frac{Q/A}{\varepsilon_0}$ for $+Q$ and $-Q$ disks; $\left|\vec{E}_{fringe}\right| \approx \frac{Q/A}{\varepsilon_0}\left(\frac{s}{2R}\right)$ just outside capacitor

$\left|\vec{E}_{dipole}\right| \approx \frac{1}{4\pi\varepsilon_0}\frac{2qs}{x^3}$ along dipole axis, for $x >> s$; $\left|\vec{E}_{dipole}\right| \approx \frac{1}{4\pi\varepsilon_0}\frac{qs}{y^3}$ along $\perp$ axis, for $y >> s$

$i = nA\bar{v}$ $I = |q|nA\bar{v}$ $\bar{v} = uE$ $\Delta V = \frac{q}{4\pi\varepsilon_0}\left[\frac{1}{r_f} - \frac{1}{r_i}\right]$ due to point charge

$\Delta\vec{B} = \frac{\mu_0}{4\pi}\frac{I\Delta\vec{l} \times \hat{r}}{r^2}$ for a short segment of current-carrying conductor $\Delta\vec{F} = I\Delta\vec{l} \times \vec{B}$

$\left|\vec{B}_{wire}\right| = \frac{\mu_0}{4\pi}\frac{LI}{r\sqrt{r^2 + (L/2)^2}} \approx \frac{\mu_0}{4\pi}\frac{2I}{r}$ for $r << L$ $\left|\vec{B}_{solenoid}\right| = \frac{\mu_0 NI}{L}$ for $L >> R$ $\vec{B}_{point} = \frac{\mu_0}{4\pi}\frac{q\vec{v} \times \hat{r}}{r^2}$

$\left|\vec{B}_{loop}\right| = \frac{\mu_0}{4\pi}\frac{2I\pi R^2}{(z^2 + R^2)^{3/2}} \approx \frac{\mu_0}{4\pi}\frac{2I\pi R^2}{z^3}$ on axis, $z >> R$ $\mu = IA = I\pi R^2$ current loop $\left|\vec{B}_{dipole}\right| \approx \frac{\mu_0}{4\pi}\frac{2\mu}{z^3}$ on axis

$\sigma = |q|nu$ $J = \frac{I}{A} = \sigma E$ $R = \frac{L}{\sigma A}$ $I = \frac{|\Delta V|}{R}$ for an ohmic resistor (R independent of ΔV)

power $= I\Delta V$ $Q = C|\Delta V|$ $\vec{E}_{radiative} = \frac{1}{4\pi\varepsilon_0}\frac{-q\vec{a}_\perp}{c^2 r}$ $\hat{v} = \hat{E} \times \hat{B}$ $B = \frac{E}{c}$

Physical constants

$\frac{1}{4\pi\varepsilon_0} = 9\times10^9 \frac{N \cdot m^2}{C^2}$ $\varepsilon_0 = 8.85\times10^{-12}\frac{C^2}{N \cdot m^2}$ $\frac{\mu_0}{4\pi} = 1\times10^{-7}\frac{T \cdot m^2}{C \cdot m/s}$ $\mu_0 = 4\pi\times10^{-7}\frac{T \cdot m^2}{C \cdot m/s}$

$c = 3\times10^8$ m/s $e = 1.6\times10^{-19}$ C $m_{proton} \approx m_{neutron} = 1.7\times10^{-27}$ kg $m_{electron} = 9\times10^{-31}$ kg

6×10^{23} molecules/mole Atomic radius $\approx 10^{-10}$ m Proton radius $\approx 10^{-15}$ m $g = 9.8$ N/kg

Electric field necessary to ionize air, about 3×10^6 N/C $B_{Earth} \approx 2\times10^{-5}$ T (horizontal component)